

SEMESTER END / BACKLOG SUBJECT EXAMINATIONS - JANUARY 2026

Program	: B.E. - Common to all Programs	Semester	: V
Course Name	: Environmental Studies	Max. Marks	: 50
Course Code	: 21HS59 / 22 / 23 / HS510	Duration	: 2 Hrs

Instructions to the Candidates:

- **PART-A** : Answer all the questions.
- **PART-B** : Answer one full question from each unit.

PART- A

CO1- (20)
CO5

- The multidisciplinary nature of Environmental Studies means:
 - It includes only biological sciences
 - It integrates physical, chemical, biological and social sciences
 - It deals only with pollution
 - It focuses only on biodiversity
- The transfer of energy from one trophic level to the next is called:
 - Bioaccumulation
 - Energy Pyramid
 - Energy Flow
 - Material Cycling
- The variety of genes, species and ecosystems is referred to as:
 - Ecology
 - Biodiversity
 - Sustainability
 - Conservation
- Sustainable development emphasizes:
 - Using resources without concern for future generations
 - Maximizing industrial growth
 - Meeting present needs without compromising future needs
 - Total ban on resource use
- The highest proportion of global freshwater is stored in:
 - Rivers
 - Glaciers and ice caps
 - Lakes
 - Groundwater
- Soil erosion occurs mainly due to:
 - Afforestation
 - Heavy rainfall and deforestation
 - Organic farming
 - Mulching
- Minerals are considered a non-renewable resource because:
 - They are biodegradable
 - They can be recycled instantly
 - Their formation takes millions of years
 - They are unlimited in reserves
- Desertification is mainly caused by:
 - Flooding
 - Overgrazing and deforestation
 - Industrialization
 - Organic farming
- Biogas mainly contains:
 - Carbon dioxide
 - Methane
 - Hydrogen
 - Nitrogen

10. The device that converts chemical energy of hydrogen into electricity is:
A) Solar panel
B) Bioreactor
C) Hydrogen fuel cell
D) Transformer
11. The main objective of Environmental Impact Assessment (EIA) is:
A) To promote mining activities
B) To identify, predict and evaluate environmental impacts
C) To increase industrial waste
D) To eliminate renewable energy
12. Nuclear energy is produced through:
A) Photosynthesis
B) Fission and fusion
C) Combustion of biomass
D) Oxidation reactions
13. Excess growth of algae due to nutrient pollution causes:
A) Biomagnification
B) Eutrophication
C) Ozone depletion
D) Acid rain
14. The major pollutant responsible for acid rain is:
A) CO₂
B) CH₄
C) SO₂ and NO_x
D) CFCs
15. Which of the following is an effect of air pollution?
A) Ozone formation at ground level
B) Increase in biodiversity
C) Soil fertility improvement
D) Noise reduction
16. The major pollutant responsible for acid rain is:
A) CO₂
B) CH₄
C) SO₂ and NO_x
D) CFCs
17. Greenhouse effect is caused due to trapping of:
A) Visible light
B) Infrared radiation
C) Gamma rays
D) UV-C rays
18. The Environmental Protection Act was enacted in:
A) 1974
B) 1986
C) 1981
D) 1971
19. Air quality standards in India are regulated by:
A) WHO
B) FAO
C) Central Pollution Control Board (CPCB)
D) ISRO
20. Which phenomenon is responsible for global warming?
A) Plate tectonics
B) Enhanced greenhouse effect
C) Ozone formation
D) Condensation

PART-B UNIT-I

1. With neat diagram, illustrate energy flow and material flow in Ecosystem. CO1 (06)
2. Describe biodiversity, its levels and various threats affecting global biodiversity. CO1 (06)

UNIT-II

3. Explain soil erosion, its causes, and the measures used to control it. CO2 (06)
4. Enumerate environmental impacts of over extraction of mineral resources. CO2 (06)

UNIT-III

5. Compare renewable and non-renewable energy sources with suitable examples. CO3 (06)
6. Describe the step-by-step procedure of conducting Environmental Impact Assessment. CO3 (06)

UNIT-IV

7. Discuss causes, effects and control measures of water pollution. CO4 (06)
8. Explain the methods used for management and treatment of municipal solid waste. CO4 (06)

UNIT-V

9. Explain global warming and climate change. Discuss their environmental impacts. CO5 (06)
10. Describe the key features of the Environmental Protection Act (1986), Air Act and Water Act. CO5 (06)
